

Pavan Kumar Gannaju

+91 6300422984 | pavan9542644804@gmail.com | [LinkedIn](#) | [Portfolio](#) | [Github](#)

PROFESSIONAL SUMMARY

AI security and backend engineer. Built a fail-closed LLM guardrail gateway whose per-class evaluation exposed a harmful-content blind spot that aggregate metrics hid, found and fixed an authorization bypass in my own design, and applied the same supply-chain discipline to Kubernetes with an admission controller for signed images and safetensors model artifacts. Every figure comes from a committed evaluation with confidence intervals or a CI job.

EXPERIENCE

AI Intern → Research Team Lead — Prodigal AI Technologies Pvt. Ltd., Delhi *Mar 2025 – Nov 2025 | Remote, part-time*

- Promoted from AI Intern to Research Team Lead; coordinated AI security R&D and authored two internal manuscripts — zero-shot voice cloning via Mixture-of-Experts architectures, and AI tamper-proofing / adversarial defence.
- Directed architecture reviews and research sprints for a small engineering team building modular, role-aware AI workflow systems; applied prompt-injection defence and role-aware execution modelling to internal infrastructure.

Gen AI / LLM Intern — Digital Nexus AI, Jersey City, NJ, USA

May 2025 – Sep 2025 | Remote, part-time

- Built backend REST APIs in Python and FastAPI supporting AI workflow systems and inter-service communication.
- Supported integration testing and validation of AI-integrated backend services ahead of release cycles.

PROJECTS

Gatekeeper — Neuro-Symbolic AI Governance Gateway (Solo) | *Python, FastAPI, FAISS, spaCy, PyTorch, Docker*

- Built a fail-closed guardrail gateway combining symbolic rules (regex, NER) with a learned fusion of 8 detectors, evaluated on a 6,933-prompt, seven-source suite with bootstrapped confidence intervals on every reported figure.
- Per-class evaluation showed the best single detector reached 0.949 AUC yet scored only 2.0% on harmful-content prompts, a blind spot the pooled metric hid. Fusion reached 0.952 AUC / 86.1% recall at 5% FPR, and fusion plus judge reached 62.6% on harmful-content offline.
- Wired the fusion into the live pipeline, raising recall from 30.0% to 63.6% (F1 0.44 → 0.71) on the same 546 prompts at the cost of a higher false-positive rate (3.8% → 8.8%); separately diagnosed a semantic-cache flaw where 9.1% of near-duplicate pairs above the shipped threshold carried opposite labels.
- Found and fixed an authorization bypass: a client-supplied role field could set INTERNAL and disable every guardrail; capability is now resolved server-side and the field is rejected with a 422. 1,644 tests; CI runs dependency audit and Trivy/SBOM image scanning.

ModelGate — Kubernetes Admission Control for ML Workloads | *Go, Kubernetes, controller-runtime, cosign, envtest*

- Built an admission webhook that admits only cosign-signed images and hash-verified safetensors model artifacts and denies privileged and host-access pods; tested both failurePolicy modes by killing the webhook.
- Adversarial testing found two real bypasses (a post-admission image swap and ephemeral containers), both fixed and covered by tests; fuzzing the pickle detector broke my first two fixes before a position-based check held.
- CI runs unit tests, envtest against a real API server, real cosign sign/verify, fuzzing and a kind-cluster smoke test.

SentinAI — Secure AI Desktop Orchestration | *Team Lead, 5 members; B.Tech thesis | Python, FastAPI, Ollama, SQLite*

- Designed an architecture that treats the LLM as untrusted: allowlist, filesystem sandbox, keyword filter and human confirmation enforced outside the model, with the LLM removed from iterative control of OS actions; blocked 66/66 cases in an adversarial fuzz suite (shell injection, path traversal, forbidden intents, random noise).
- Measured 96.7% end-to-end task success (95% CI 91.7–98.7%, n=120) by verifying real OS state rather than the agent's own report; the intent router reaches 92.33% on real-world phrasing and 83.68% out-of-distribution, with 88.45% of requests resolved without a model call. 990 tests, 87.5% coverage.

Agentic-OS — Capability-Based x86_64 UEFI Kernel (Solo) | *Rust (no_std), UEFI, QEMU, GitHub Actions*

- Wrote a kernel where all access goes through per-process capabilities with attenuation on derive and O(1) generation-counter revocation; Linux CI boots it in QEMU and asserts that revocation is enforced, and fuzzes the ext2 parsers. Runs under QEMU only.

SKILLS

Languages: Python, Go, Rust, Java, C/C++, SQL (PostgreSQL, MySQL)

AI Security: Prompt-injection defence, adversarial testing, OWASP LLM Top 10, guardrail design, detector fusion, agent sandboxing, capability-based authorization, supply-chain verification (cosign, SBOM, Trivy)

AI & Generative AI: LLMs, RAG, vector search (FAISS), embeddings, HuggingFace Transformers, spaCy NER, scikit-learn

Backend & Infrastructure: FastAPI, asyncio, REST APIs, Kubernetes (kind, controller-runtime), Docker, GitHub Actions, AWS, Linux

Evaluation: pytest, bootstrapped and Wilson confidence intervals, adversarial fuzzing, envtest, coverage-gated CI

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, Distributed Systems

EDUCATION

B.Tech, Computer Science & Engineering (AI/ML) — Visvesvaraya College of Engineering and Technology (JNTUH) | April 2026 | CGPA: 8.01/10.0

IIT Ropar – Major in Artificial Intelligence | online programme via Masai/NSDC | 2024 – 2026

RESEARCH

Two unpublished manuscripts authored during industry R&D at Prodigal AI: expressivity in zero-shot voice cloning via Mixture-of-Experts architectures, and a defence-in-depth framework against prompt injection, data poisoning and adversarial manipulation of LLMs.

CODING & LEADERSHIP

[LeetCode](#) 530+ solved including 86 Hard · [Codeforces](#) 249 problems solved (218 in 2026) · Organised and mentored two university hackathons (50+ participants), providing debugging and algorithmic problem-solving support